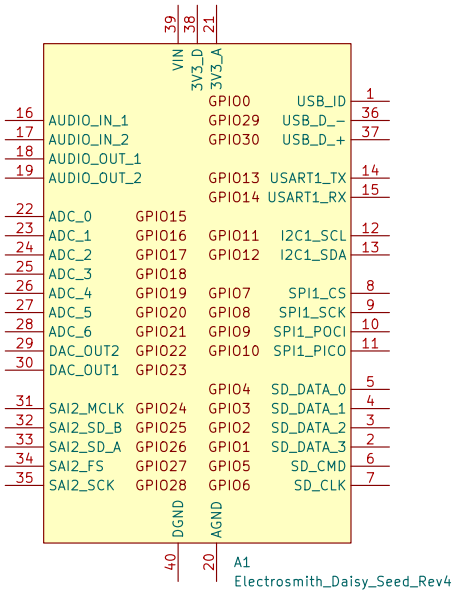
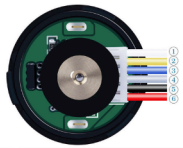


My plan is to remove the dupont housing from the motor, while keeping the crimped ends in place.
I will then secure the motor wires to the screw terminals on the driver (maybe via some crimped termination) and refit the encoder cables to a 4 slot dupont that will have a 3D−printed attachment to secure it.

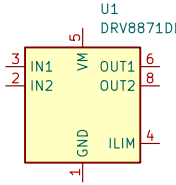
I am also interested in buying snapfit sockets and terminations. We will just need to see what works best.

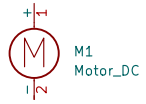
The motor/encoder has six color-coded, 8" (20 cm) leads terminated by a 1×6 female header with a 0.1" pitch, as shown in the main product picture. This header works with standard 0.1" male headers and our male jumper and precrimped wires. If this header is not convenient for your application, you can pull the crimped wires out of the header or cut the header off. The following table describes the wire functions:

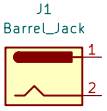


White	Encoder B Output
Yellow	Encoder A Output
Blue	Encoder VCC (3.3V to 24V V)
Gray	Encoder GND
Black	Motor Power (Connects to the Other Motor Terminal)
Red	Motor Power (Connects to One Motor Terminal)

The Hall sensor requires an input voltage, Vcc, between 3.3 and 24 V and draws a maximum of 10 mA. The A and B outputs are square waves from 0 V to Vcc approximately 90° out of phase. The frequency of the transitions tells you the speed of the motor, and the order of the transitions tells you the direction. The following oscilloscope capture shows the A and B (yellow and white) encoder outputs using a motor voltage of 6 V and a Hall sensor Vcc of 5 V:







Sheet: /
File: SlidingGuitar_Schematic.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 1/1